

Normalization – Airline Booking System

1NF (Payment)

Dependencies

Dependency Type	Functional Dependency
Full Dependency	PaymentID → Amount, PaymentStatus, PaymentMode
Partial Dependency	None
Transitive Dependency	PaymentID → BookingID → SeatNo, TravelClass, BookingStatus, PassengerId, FirstName, LastName, Gender, DOB, Phone, Email, PassportNo, Nationality, UserId, UserName, Password, Role, ScheduleId, DepartureTime, ArrivalTime, TotalSeats, FlightStatus, FlightId, FlightNo, Duration, BasePrice, AirportId, AirportName, IATACode, City, Country, AirlineId, AirlineName, IATACode, Country, ContactNo, AirlineEmail

Normalization Result

Normal Form	Result
2NF	Not possible (no partial dependency)
3NF Tables	Payment

1NF (Booking)

Dependencies

Dependency Type	Functional Dependency
Full Dependency	BookingID → SeatNo, TravelClass, BookingStatus
Partial Dependency	None
Transitive Dependency	BookingID → PassengerID → FirstName, LastName, Gender, DOB, Phone, Email, PassportNo, Nationality BookingID → UserID → UserName, Password, Role BookingID → ScheduleId → DepartureTime, ArrivalTime, TotalSeats, FlightStatus, FlightId, FlightNo, Duration, BasePrice, AirportId, AirportName, IATACode, City, Country, AirlineId, AirlineName, IATACode, Country, ContactNo, AirlineEmail

Normalization Result

Normal Form	Result
2NF	Not possible (no partial dependency)
3NF Tables	Booking, Passenger, User

1NF (Schedule)

Dependencies

Dependency Type	Functional Dependency
Full Dependency	ScheduleId → DepartureTime, ArrivalTime, TotalSeats, FlightStatus
Partial Dependency	None
Transitive Dependency	ScheduleId → FlightId → FlightNo, Duration, BasePrice, AirlineId, AirportId FlightId → AirlineId → AirlineName, IATACode, Country, ContactNo, AirlineEmail FlightId → AirportId → AirportName, IATACode, City, Country

Normalization Result

Normal Form	Result
2NF	Not possible (no partial dependency)
3NF Tables	Schedule

1NF (Flight)

Dependencies

Dependency Type	Functional Dependency
Full Dependency	FlightId → FlightNo, Duration, BasePrice
Partial Dependency	None
Transitive Dependency	FlightId → AirlineId → AirlineName, IATACode, Country, ContactNo, AirlineEmail FlightId → SourceAirportId → SourceAirportName, SourceIATACode, SourceCity, SourceCountry FlightId → DestinationAirportId → DestinationAirportName, DestinationIATACode, DestinationCity, DestinationCountry

Normalization Result

Normal Form	Result
2NF	Not possible (no partial dependency)
3NF Tables	Flight, Airline, SourceAirport, DestinationAirport

Final Summary of 3NF Tables

Final Tables
1. Payment
2. Booking
3. Passenger
4. User
5. Schedule
6. Flight
7. Airline
8. SourceAirport
9. DestinationAirport